Quantum Derivation of Schottky Diode Current

Coupled Poisson–Schrödinger–Landauer Framework

Nomenclature

Symbol	Definition	SI unit
A^{**}	Effective Richardson constant	A m^{-2} K^{-2}
E, E_x	Total and longitudinal electron energy	J
E_C, E_F	Conduction-band edge and Fermi level	J
F, F_s	Electric field and surface field	V m^{-1}
f_S, f_M	Fermi occupations	–
$h, \hbar$	Planck constants	J s
J, J_S	Current and saturation-current density	A m^{-2}
k_B, T	Boltzmann constant and temperature	J K^{-1}, K
m^*, m_0	Effective and free-electron mass	kg
$M(E)$	Propagating modes per unit area	m^{-2}
N_D	Donor concentration	m^{-3}
q	Positive elementary charge	C
$U(x)$	Electron potential energy	J
V, V_d	Applied voltage and depletion potential	V
W	Depletion width	m
ε_s	Semiconductor permittivity	F m^{-1}
$\Phi_B, \Delta\Phi_B$	Barrier height and lowering	V
$\mathcal{T}(E)$	Transmission probability	–
ψ, Ψ	Electrostatic potential and wavefunction	V, –

Convention. $q > 0$; electron charge is $-q$. Φ_B is in volts, so barrier energy is $q\Phi_B$.

1 Poisson depletion solution

For uniform ionized donors, $d^2\psi/dx^2 = -qN_D/\varepsilon_s$. With $F = -d\psi/dx$ and $F(W) = 0$,

$$F(x) = \frac{qN_D}{\varepsilon_s}(W - x), \quad F_s = \frac{qN_D W}{\varepsilon_s}, \quad V_d = \frac{qN_D W^2}{2\varepsilon_s}.$$

Thus $W = \sqrt{2\varepsilon_s V_d/(qN_D)}$ and $F_s = \sqrt{2qN_D V_d/\varepsilon_s}$.

2 Image-force lowering

For a planar interface and locally constant field,

$$U(x) = q\Phi_B - qF_s x - \frac{q^2}{16\pi\varepsilon_s x}.$$

Setting $dU/dx = 0$ gives $x_m = \sqrt{q/(16\pi\varepsilon_s F_s)}$ and

$$\Delta\Phi_B = \sqrt{\frac{qF_s}{4\pi\varepsilon_s}}, \qquad \Phi_{B,\text{eff}} = \Phi_B - \Delta\Phi_B.$$

The continuum image potential is not valid arbitrarily close to the atomic interface.

3 Schrödinger transmission

$$-\frac{\hbar^2}{2m^*}\frac{d^2\Psi}{dx^2} + U(x)\Psi = E_x\Psi.$$

For $U = U_0 - qFx$, define $\ell = [\hbar^2/(2m^*qF)]^{1/3}$ and $z = (x_t - x)/\ell$. The equation becomes $\Psi_{zz} - z\Psi = 0$, with solutions $\mathrm{Ai}(z)$ and $\mathrm{Bi}(z)$. For a smooth forbidden region,

$$\mathcal{T}(E_x) \approx \exp\left[-\frac{2}{\hbar}\int_{x_1}^{x_2}\sqrt{2m^*(U - E_x)}\,dx\right].$$

For a triangular barrier this reduces to

$$\mathcal{T}(E_x) \approx \exp\left[-\frac{4\sqrt{2m^*}(U_0 - E_x)^{3/2}}{3\hbar qF}\right].$$

4 Richardson law from state counting

For a 3D parabolic band with spin degeneracy two,

$$\Gamma_x = \frac{2}{(2\pi)^3}\int_{k_x>0}\frac{\hbar k_x}{m^*}f(E)\,d^3k.$$

Using nondegenerate statistics and a longitudinal threshold gives

$$A^{**} = \frac{4\pi q m^* k_B^2}{h^3}, \qquad J_S = A^{**}T^2 e^{-q\Phi_B/(k_BT)}.$$

Here Φ_B is referenced from the semiconductor Fermi level; if using $E_C - E_F$ separately, do not count it twice.

5 Landauer current and ideal limit

$$J = \frac{q}{h}\int M(E)\mathcal{T}(E,V)[f_S(E) - f_M(E)]\,dE.$$

With ideal thermionic transmission and detailed balance,

$$J = J_S\left[e^{qV/(k_BT)} - 1\right], \qquad J_S = A^{**}T^2 e^{-q\Phi_B/(k_BT)}.$$

This is the ideal $n = 1$ limit. A quantitative model requires consistent mode normalization, electrochemical potentials, and a bias-dependent barrier.

6 Numerical workflow

1. Specify material parameters, doping, temperature, barrier, and boundary conditions.
2. Solve Poisson for potential and field; update carrier statistics if needed.
3. Construct the barrier and state any image-potential regularization.
4. Solve the scattering problem for $\mathcal{T}(E)$ and verify convergence/current conservation.
5. Integrate occupied modes with the two reservoir Fermi functions.
6. Sweep voltage and temperature; verify equilibrium detailed balance and the Richardson limit.

7 Limitations

Parabolic isotropic effective mass, depletion approximation, ideal interface, and WKB smooth-barrier assumptions limit the closed forms. Si, 4H-SiC, and GaN require material-specific masses, valley structure, interface properties, and barrier data for quantitative predictions.

References

R. T. Tung, Applied Physics Reviews **1**, 011304 (2014), DOI: 10.1063/1.4858400.
S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 3rd ed., Wiley (2007).